

(19)



(11)

EP 3 221 308 B1

(12)

EUROPEAN PATENT SPECIFICATION

(45) Date of publication and mention
of the grant of the patent:
19.09.2018 Bulletin 2018/38

(51) Int Cl.:
C07D 403/14 ^(2006.01) **A61K 31/506** ^(2006.01)
A01N 43/60 ^(2006.01) **A61P 31/10** ^(2006.01)

(21) Application number: **15800925.8**

(86) International application number:
PCT/GB2015/053546

(22) Date of filing: **20.11.2015**

(87) International publication number:
WO 2016/079536 (26.05.2016 Gazette 2016/21)

(54) **ANTIFUNGAL AGENTS**

ANTIMYKOTIKA

AGENTS ANTIFONGIQUES

(84) Designated Contracting States:
**AL AT BE BG CH CY CZ DE DK EE ES FI FR GB
GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO
PL PT RO RS SE SI SK SM TR**

(30) Priority: **21.11.2014 GB 201420743**
07.04.2015 GB 201505891

(43) Date of publication of application:
27.09.2017 Bulletin 2017/39

(60) Divisional application:
18186872.0

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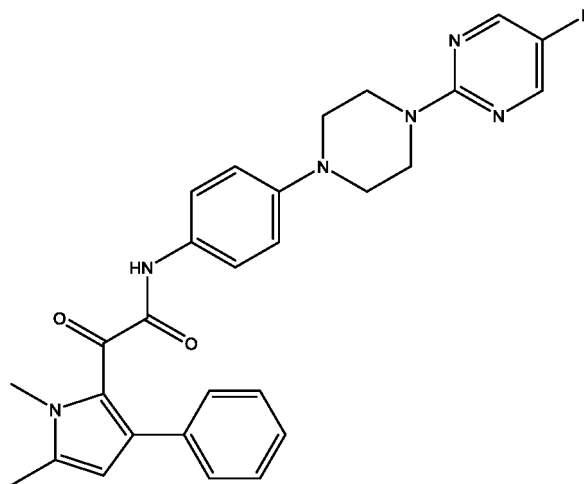
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WO-A1-2009/130481 WO-A1-2009/144473

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Figure 7 shows the cumulative mortality of immunosuppressed mice infected with *L. proliferans* FMR 3569 when treated with (a) VRC, voriconazole at 25mg/kg p.o. by gavage QD; (b) Example 1 at 20mg/kg, p.o. by gavage BID and (c) no treatment.

Detailed Description of the Invention

[0028] The invention provides a compound, which compound is



2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide or a pharmaceutically acceptable salt thereof.

[0029] For the avoidance of doubt, the compound can, if desired, be used in the form of solvates. Further, for the avoidance of doubt, the compound may be used in any tautomeric form.

[0030] As used herein, a pharmaceutically acceptable salt is a salt with a pharmaceutically acceptable acid or base. Pharmaceutically acceptable acids include both inorganic acids such as hydrochloric, sulphuric, phosphoric, diphosphoric, hydrobromic, hydroiodic or nitric acid and organic acids such as citric, fumaric, maleic, malic, ascorbic, succinic, tartaric, benzoic, acetic, methanesulphonic, ethanesulphonic, benzenesulphonic, p-toluenesulphonic acid, formic, acetic, propionic, glycolic, lactic, pyruvic, oxalic, salicylic, trichloroacetic, picric, trifluoroacetic, cinnamic, pamoic, malonic, mandelic, bismethylene salicylic, ethanedisulfonic, gluconic, citraconic, aspartic, stearic, palmitic, EDTA, p-aminobenzoic or glutamic acid, sulfates, nitrates, phosphates, perchlorates, borates, acetates, benzoates, hydroxynaphthoates, glycerophosphates or ketoglutarates. Further examples of pharmaceutically acceptable inorganic or organic acid addition salts include the pharmaceutically acceptable salts listed in Journal of Pharmaceutical Science, 66, 2 (1977) which are known to the skilled artisan. Pharmaceutically acceptable bases include alkali metal (e.g. sodium or potassium) and alkali earth metal (e.g. calcium or magnesium) hydroxides and organic bases such as alkyl amines, aralkyl amines and heterocyclic amines, lysine, guanidine, diethanolamine and choline.

[0031] Also intended as pharmaceutically acceptable acid addition salts are the hydrates which the present compound is able to form.

[0032] The acid addition salts may be obtained as the direct products of compound synthesis. In the alternative, the free base may be dissolved in a suitable solvent containing the appropriate acid, and the salt isolated by evaporating the solvent or otherwise separating the salt and solvent.

[0033] The compound may form solvates with standard low molecular weight solvents using methods known to the skilled artisan.

Prodrug

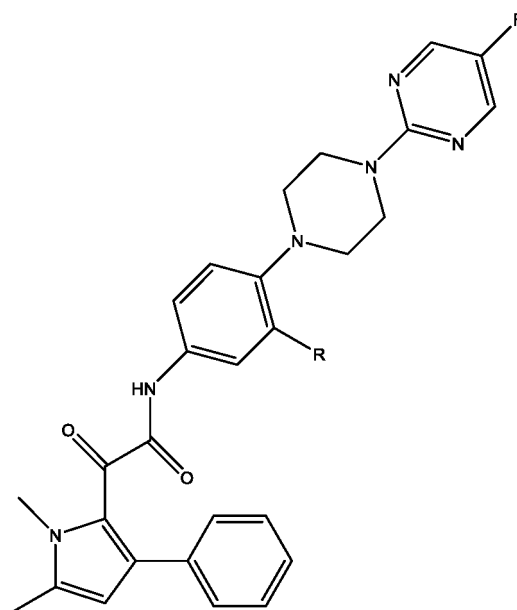
[0034] It is possible to form a prodrug of 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide. The prodrug may, for instance, provide enhanced solubility, permeability, adsorption, distribution and formulation, and/or lower toxicity.

[0035] A prodrug is an analogue of the compound of the invention which will be converted *in vivo* to the desired active compound. Suitable methods will be known to those skilled in the art.

[0036] Particularly suitable prodrugs include those in which a nitrogen atom of the compound is quaternised by addition of an ester or alkyl ester group. For example, the nitrogen atom of an amine group may be quaternised by addition of a

-CH₂-O-COR group, wherein R is typically methyl or tert-butyl.

[0037] Other suitable prodrugs include those in which a moiety is added to the phenyl ring adjacent to the piperazinyl group. Relative to the amide (-NH-CO-) moiety, the moiety may be added to the phenyl ring at the ortho or meta position, preferably at the meta position. The prodrug may, for instance, have the general formula:



wherein R is a group of formula -O-CO-OR¹, -O-CO-R¹, -O-CO-NR¹R², -O-CO-(CH₂)_z-NR¹R², -OR¹, -O-(CR¹R²)_z-O-CO-R³, -O-P(O)(OR⁴)(OR⁵) or -O-(CH₂)_z-O-P(O)(OR⁴)(OR⁵), wherein: R¹, R² and R³ are independently hydrogen, C1-C4 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C3-C6 cycloalkyl, an unsubstituted 5- to 7-membered heterocyclyl group, or a 5- to 7-membered heterocyclyl group substituted with up to three substituents selected from C1-C4 alkyl and C1-C4 alkoxy; R⁴ and R⁵ are independently hydrogen, C1-C4 alkyl, C2-C4 alkenyl, C2-C4 alkynyl or a group I element such as Na; and z is 1, 2, 3 or 4.

[0038] As used herein, a C1-C4 alkyl group or moiety can be linear or branched but is preferably linear. Suitable such alkyl groups and moieties include methyl, ethyl, n-propyl, i-propyl, n-butyl, sec-butyl and tert-butyl.

[0039] As used herein, a C2-C4 alkenyl group or moiety can be linear or branched but is preferably linear. It contains one or more carbon-carbon double bonds. It is preferably a C2-C3 alkenyl group. Suitable such alkenyl groups and moieties include vinyl, allyl, propenyl and butenyl, e.g. CH₂C(Me)=CH₂.

[0040] As used herein, a C2-C4 alkynyl group or moiety can be linear or branched but is preferably linear. It contains one or more carbon-carbon triple bonds. It is preferably a C2-C3 alkynyl group. Suitable alkynyl groups and moieties include ethynyl, propynyl, and butynyl, and isomers thereof.

[0041] As used herein, a C3-C6 cycloalkyl group is typically a C4, C5 or C6 cycloalkyl group, more preferably a C5 or C6 cycloalkyl group.

[0042] An alkyl, alkenyl, alkynyl or cycloalkyl group is unsubstituted.

[0043] As used herein and unless otherwise stated, a heterocyclyl group or moiety is a saturated 5- to 7-membered ring system in which the ring contains at least one heteroatom. Typically, the ring contains up to three heteroatoms, e.g. one or two heteroatoms, selected from O, S and N. Thus, a heterocyclyl group or moiety is typically a 5- to 7-membered ring containing one, two or three heteroatoms selected from O, S and N. Suitable such heterocyclyl groups and moieties include, for example, monocyclic saturated 5- to 7-membered rings, more preferably monocyclic saturated 5- to 6-membered rings such as tetrahydrofuranyl, piperidinyl, oxazolidinyl, morpholinyl, thiomorpholinyl, pyrrolidinyl, dioxolanyl, piperidonyl, piperazinyl, tetrahydropyranyl and 1,4-diazepanyl, more preferably pyrrolidinyl, piperazinyl, tetrahydropyranyl and piperidinyl.

[0044] A heterocyclyl group may be substituted or unsubstituted. Each ring atom may be unsubstituted or may carry one or two substituents. If desired, a nitrogen atom may be disubstituted and a sulphur atom may be substituted, providing a charged heteroatom. Typically, a heterocyclyl group carries up to three substituents, e.g. one or two substituents. The heterocycle may be connected to the remainder of the molecule by a bond to any of its available ring positions. Suitable substituents are C1-C4 alkyl and C1-C4 alkoxy, e.g. methyl, ethyl, methoxy and ethoxy, preferably methyl.

[0045] Preferably, R¹, R² and R³ are independently hydrogen, C1-C4 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C3-C6 cycloalkyl, or an unsubstituted 5- to 6-membered heterocyclyl group. More preferably, R¹, R² and R³ are independently

hydrogen, C1-C4 alkyl, C3-C6 cycloalkyl, or an unsubstituted pyrrolidinyl, piperazinyl, tetrahydropyranyl or piperidinyl group.

[0046] Preferably, R⁴ and R⁵ are independently hydrogen, C1-C4 alkyl, or a group I element such as Na. More preferably, R⁴ and R⁵ are independently hydrogen, methyl, ethyl, or Na.

[0047] Preferably, z is 1 or 2. More preferably z is 1.

[0048] Preferably R is a group of formula -O-CO-OR¹, -O-CO-R¹, -O-CO-NR¹R², -O-CO-(CH₂)_z-NR¹R², -OR¹, -O-(CR¹R²)_z-O-CO-R³ or -O-P(O)(OR⁴)(OR⁵). More preferably, R is a group of formula -O-COR¹, -O-CO-NR¹R², -O-CO-(CH₂)_z-NR¹R², -OR¹, -O-(CR¹R²)_z-O-CO-R³ or -O-P(O)(OR⁴)(OR⁵).

[0049] In one embodiment, R is a group of formula -O-CO-OR¹, -O-CO-R¹, -O-CO-NR¹R², -O-CO-(CH₂)_z-NR¹R², -OR¹, -O-(CR¹R²)_z-O-CO-R³ or -O-P(O)(OR⁴)(OR⁵), wherein: R¹, R² and R³ are independently hydrogen, C1-C4 alkyl, C2-C4 alkenyl, C2-C4 alkynyl, C3-C6 cycloalkyl, or an unsubstituted 5- to 6-membered heterocyclyl group; R⁴ and R⁵ are independently hydrogen, C1-C4 alkyl, or a group I element such as Na; and z is 1 or 2. More preferably R is a group of formula -O-COR¹, -O-CO-NR¹R², -O-CO-(CH₂)_z-NR¹R², -OR¹, -O-(CR¹R²)_z-O-CO-R³ or -O-P(O)(OR⁴)(OR⁵), wherein R¹, R² and R³ are independently hydrogen, C1-C4 alkyl, C3-C6 cycloalkyl, or an unsubstituted pyrrolidinyl, piperazinyl, tetrahydropyranyl or piperidinyl group; R⁴ and R⁵ are independently hydrogen, methyl, ethyl, or Na; and z is 1.

[0050] R may, for example, be -OP(O)(ONa)₂, -OP(O)(OH)₂, -OC(=O)CH₂N(H)CH₃, -OC(=O)C₄NH₈, -OC(=O)CH₃, -OC(=O)N(CH₃)₂ or -OCH₂OC(=O)C(CH₃)₃.

[0051] In one embodiment, the compound is a pharmaceutically acceptable salt of the prodrug.

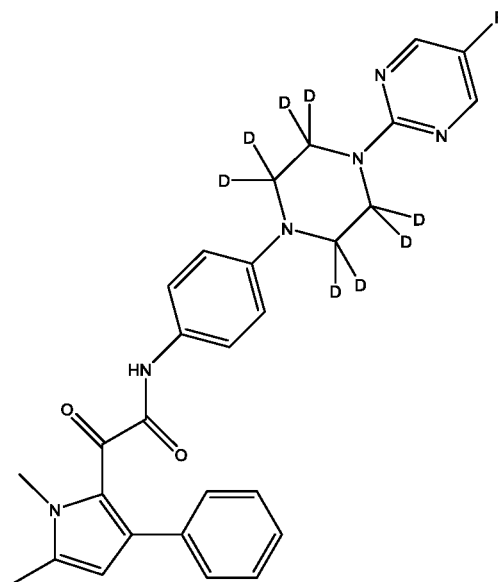
Deuterated Derivative

[0052] Also mentioned herein is a deuterated derivative of 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)-3-hydroxyphenyl)-2-oxoacetamide. Deuterated derivatives of the compound of the invention have been found to be highly active antifungal agents.

[0053] The term deuterated derivative as used herein refers to a compound in which at least one hydrogen atom has been replaced with deuterium, for instance, from 1 to 10 hydrogen atoms may be replaced with deuterium. Two or more hydrogen atoms may be replaced with deuterium. For example, all of the hydrogen atoms on a particular ring within the compound of the invention may be replaced with deuterium.

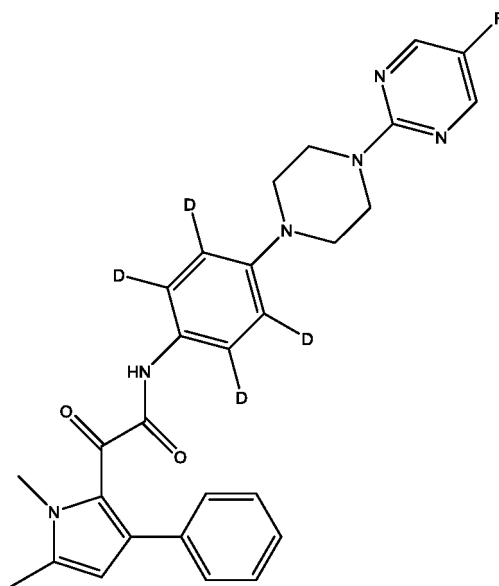
[0054] For example, a deuterated derivative of 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)-3-hydroxyphenyl)-2-oxoacetamide may comprise:

(a) all of the hydrogen atoms on the piperazinyl ring may be replaced with deuterium



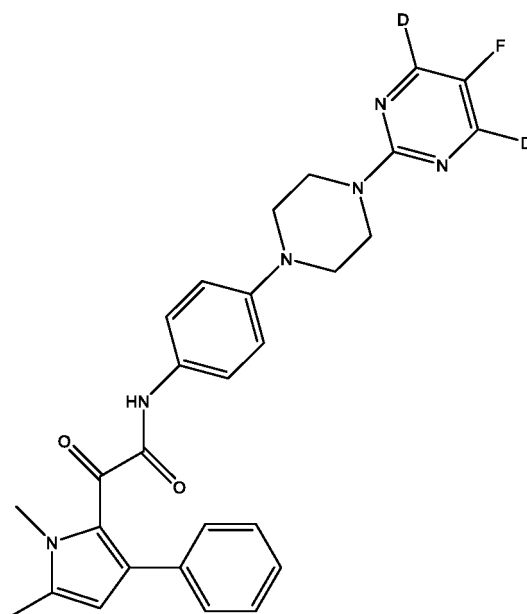
or

(b) all of the hydrogen atoms on the phenyl ring adjacent to the piperazinyl ring may be replaced with deuterium



or

(c) all of the hydrogen atoms on the pyrimidinyl ring may be replaced with deuterium



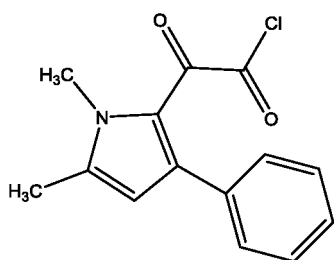
[0055] Preferably, (a) all of the hydrogen atoms on the piperazinyl ring are replaced with deuterium, or (b) all of the hydrogen atoms on the phenyl ring adjacent to the piperazinyl ring are replaced with deuterium.

[0056] Also disclosed herein are pharmaceutically acceptable salts of the deuterated derivative.

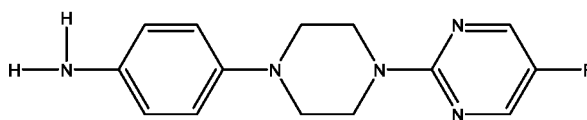
Synthesis

[0057] In one embodiment, the compound of the invention is: 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically acceptable salt thereof.

[0058] The compound may be synthesised by reacting a compound of formula (II), with a compound of formula (III). Typically the reaction takes place in the presence of an organic solvent and a base. Preferably the solvent is dichloromethane or tetrahydrofuran and the base is triethylamine or pyridine. Typically the reaction is carried out at 0°C initially while the reagents are added and then stirred at room temperature until the reaction is complete. The compound of formula (III) is typically available from commercial sources or can be prepared by known methods.

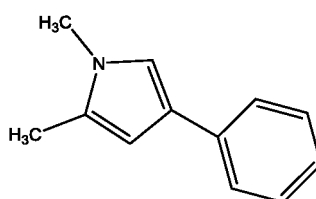


(II)



(III)

[0059] The compound of formula (II) may be prepared by reacting a compound of formula (IV), with preferably oxalyl chloride. Typically the reaction takes place in an organic solvent. Preferably, the solvent is dichloromethane. Typically, the reaction is carried out at 0 °C initially while the reagents are added and then stirred at room temperature until the reaction is complete.



(IV)

[0060] All of the starting materials referred to in the reactions described above are available from commercial sources or can be prepared by analogy with known methods.

[0061] Deuterated derivatives may be synthesised from deuterated starting materials, wherein hydrogen atoms in the starting materials are replaced with deuterium as appropriate to arrive at the target product. For example, to synthesise a deuterated derivative in which the hydrogen atoms on the piperazinyl ring are replaced with deuterium, compound (III) above may comprise deuterium in place of hydrogen on the piperazinyl ring.

[0062] The experimental section provides specific synthetic examples.

Pharmaceutical Composition

[0063] In one embodiment, the present invention provides a pharmaceutical composition comprising a pharmaceutical compound of the invention, together with one or more pharmaceutically acceptable carriers and/or excipients.

[0064] Thus the composition may comprise a compound, which compound is 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide or a pharmaceutically acceptable salt thereof, together with one or more pharmaceutically acceptable carriers and/or excipients.

[0065] The pharmaceutical composition of the invention typically contains up to 85 wt% of the pharmaceutical compound (2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide or a pharmaceutically acceptable salt thereof). More typically, it contains up to 50 wt% of the pharmaceutical compound (2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide or a pharmaceutically acceptable salt thereof). Preferred pharmaceutical compositions are sterile and pyrogen free.

Pharmaceutical Combination

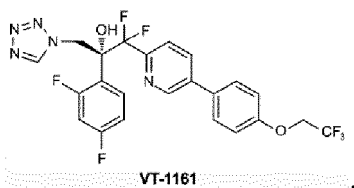
[0066] The present invention also provides a pharmaceutical combination comprising: (i) a compound which is 2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically acceptable salt thereof; and (ii) a second antifungal agent.

[0067] Typically, the pharmaceutical combination is a pharmaceutical combination in which pharmaceutical compound (i) and second antifungal agent (ii) are formulated for separate, simultaneous or successive administration. For simultaneous administration, (i) and (ii) may for example be provided in a single composition. For separate or successive administration, (i) and (ii) may, for example, be provided as a kit.

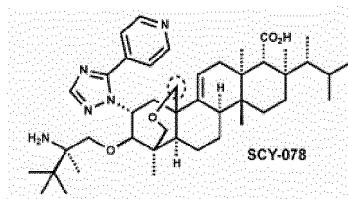
[0068] The second antifungal agent used in the invention can be any suitable antifungal agent that the skilled person

would judge to be useful in the circumstances. Particularly suitable classes of antifungal agents include azoles, polyenes, purine nucleotide inhibitors, pyrimidine nucleotide inhibitors, mannan inhibitors, protein elongation factor inhibitors, chitin synthase inhibitors, Beta-glucan synthase inhibitors, echinocandins, allylamines, anti-HSP90 antibodies, bactericidal/permeability inducing protein products and polyoxins. Other suitable antifungal agents which do not fall within the classes above include the compounds 5-fluoro-1,3-dihydro-1-hydroxy-2,1-benzoxaborale (AN269), 5-chloro-1,3-dihydro-1-hydroxy-2,1-benzoxaborale (AN2718) and icofungipen. For instance, the second antifungal agent may be selected from the group consisting of azoles, polyenes, purine nucleotide inhibitors, pyrimidine nucleotide inhibitors, mannan inhibitors, protein elongation factor inhibitors, echinocandins, allylamines, anti-HSP90 antibodies, bactericidal/permeability inducing protein products or polyoxins, or one of the compounds 5-fluoro-1,3-dihydro-1-hydroxy-2,1-benzoxaborale (AN269), 5-chloro-1,3-dihydro-1-hydroxy-2,1-benzoxaborale (AN2718), icofungipen, VT116 or SCY078.

[0069] VT116 is 2-Pyridineethanol, α -(2,4-difluorophenyl)- β , β -difluoro- α -(1*H*-tetrazol-1-ylmethyl)-5-[4-(2,2,2-trifluoroethoxy)phenyl]-, (α R)-,



and SCY078 078 (aka MK-3118) is a semi-synthetic derivative of enfumafungin, 4*H*-1,4a-Propano-2*H*-phenanthro[1,2-*c*]pyran-7-carboxylic acid, 15-[(2*R*)-2-amino-2,3,3-trimethylbutoxy]-8-[(1*R*)-1,2-dimethylpropyl]-1,6,6a,7,8,9,10,10a,10b,11,12,12a-dodecahydro-1,6a,8,10a-tetramethyl-14-[5-(4-pyridinyl)-1*H*-1,2,4-triazol-1-yl]-, (1*S*,4*aR*,6*aS*,7*R*,8*R*,10*aR*,10*bR*,12*aR*,14*R*,15*R*):



[0070] Preferred azoles are clotrimazole, econazole, bifonazole, butoconazole, fenticonazole, fluconazole, isoconazole, itraconazole, ketoconazole, miconazole, oxiconazole, sertaconazole, sulconazole, tioconazole, isavuconazole, ravuconazole, posaconazole, terconazole and voriconazole, luliconazole. Preferred echinocandins are anidulafungin, caspofungin micafungin and bialafungin. Preferred allylamines are terbinafine, butenafine, amorolfine and naftifine. Preferred polyenes are amphotericin B and nystatin. A preferred example of a purine or pyrimidine nucleotide inhibitor is flucytosine. A preferred mannan inhibitor is pradamicin. A preferred protein elongation factor inhibitor is sordarin and analogues thereof. A preferred polyoxin is nikkomycin Z.

[0071] Particularly preferred second antifungal agents are caspofungin, micafungin, anidulofungin, amphotericin B, voriconazole, posaconazole, isavuconazole, fluconazole and itraconazole.

[0072] The pharmaceutical combination may be formulated as a single composition. Thus, the pharmaceutical composition may, for example, comprise (i) a pharmaceutical compound of the invention (e.g. 2-(1,5-dimethyl-3-phenyl-1*H*-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically acceptable salt thereof), (ii) a second antifungal agent as defined herein, and (iii) a pharmaceutically acceptable carrier or diluent.

Medical Uses

[0073] In one embodiment, (i) a pharmaceutical compound of the invention, 2-(1,5-dimethyl-3-phenyl-1*H*-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically acceptable salt thereof, (ii) a pharmaceutical composition as defined herein or (iii) a pharmaceutical combination as defined herein, may be for use in a method of treatment of the human or animal body by therapy.

[0074] Accordingly, (i) a pharmaceutical compound of the invention, (ii) a pharmaceutical composition as defined herein or (iii) a pharmaceutical combination as defined herein, may be for use in the prevention or treatment of fungal disease, for example, (i) or (ii) may be used in combination with a second antifungal agent. In particular, 2-(1,5-dimethyl-3-phenyl-1*H*-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically

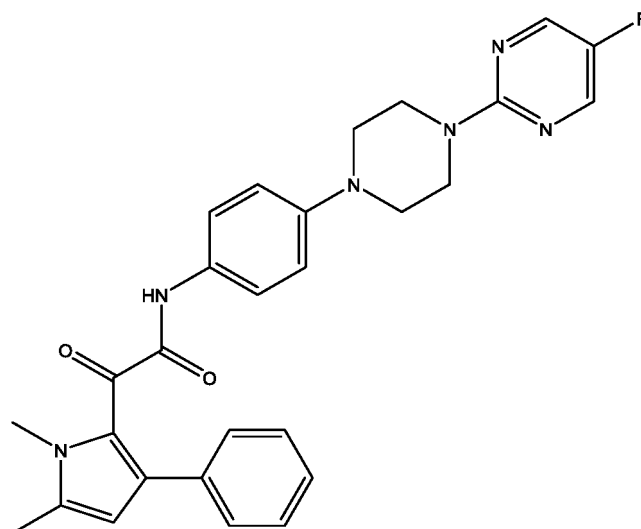
Tissue burden statistics

[0198]

Organ	Lung (control)	Kidney (control)	Brain (control)
Lung (VRC 25 mg/kg)	0.4	-	-
Kidney (VRC 25 mg/kg)	-	> 0.99	-
Brain (VRC 25 mg/kg)	-	-	0.62
Lung (Example 1 20 mg/k BID)	0.0396	-	-
Kidney (Example 1 20 mg/k BID)	-	0.07	-
Brain (Example 1 20 mg/k BID)	-	-	0.13
	Lung (VRC 25 mg/k)	Kidney (VRC 25 mg/k)	Brain (VRC 25 mg/k)
Lung (Example 1 20 mg/k BID)	0.22	-	-
Kidney (Example 1 20 mg/k BID)	-	0.01	-
Brain (Example 1 20 mg/k BID)	-	-	0.0412

Claims

1. A compound, which compound is:



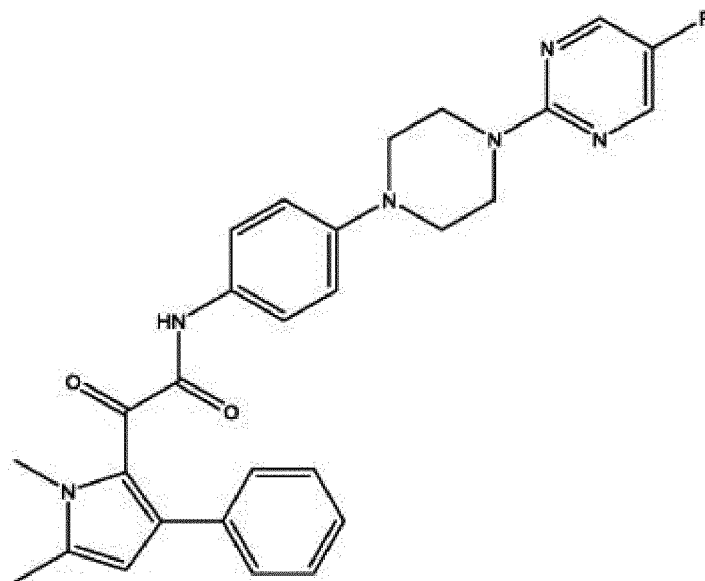
2-(1,5-dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamide, or a pharmaceutically acceptable salt thereof.

2. A pharmaceutical composition comprising a compound as defined in claim 1 together with one or more pharmaceutically acceptable carriers and/or excipients.

3. A pharmaceutical combination comprising:

- (i) a compound as defined in claim 1; and
- (ii) a second antifungal agent.

4. A pharmaceutical combination according to claim 3, wherein the compound (i) and the second antifungal agent (ii) are formulated for separate, simultaneous or successive administration.



2-(1,5-Dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamid, oder
oder ein pharmazeutisch annehmbares Salz davon.

2. Pharmazeutische Zusammensetzung, umfassend eine Verbindung wie definiert in Anspruch 1 zusammen mit einem oder mehreren pharmazeutisch annehmbaren Trägern und/oder Hilfsstoffen.
3. Pharmazeutische Kombination, umfassend:
 - (i) eine Verbindung wie definiert in Anspruch 1; und
 - (ii) ein zweites antimykotisches Mittel.
4. Pharmazeutische Kombination nach Anspruch 3, wobei die Verbindung (i) und das zweite antimykotische Mittel (ii) für eine separate, gleichzeitige oder aufeinanderfolgende Verabreichung formuliert sind.
5. Pharmazeutische Kombination nach Anspruch 3 oder Anspruch 4, wobei das zweite antimykotische Mittel ausgewählt ist aus der Gruppe bestehend aus Azolen, Polyenen, Purinnukleotid-Hemmern, Pyrimidinnukleotid-Hemmern, Mannan-Hemmern, Proteinelongationsfaktor-Hemmern, Echinocandinen, Allylaminen, Antikörpern anti-HSP90, bakterienabtötende/Permeabilität-induzierende Proteinprodukte oder Polyoxine, oder einer der Verbindungen 5-Fluor-1,3-dihydro-1-hydroxy-2,1-benzoxaboral, 5-Chlor-1,3-dihydro-1-hydroxy-2,1-benzoxaboral, Icofungipen, VT116 oder SCY078.
6. Pharmazeutische Kombination nach Anspruch 5, wobei das zweite antimykotische Mittel (a.i) ein Azol ist, ausgewählt aus Clotrimazol, Econazol, Bifonazol, Butoconazol, Fenticonazol, Fluconazol, Isoconazol, Itraconazol, Ketoconazol, Miconazol, Oxiconazol, Sertaconazol, Sulconazol, Tioconazol, Isavuconazol, Ravuconazol, Posaconazol, Terconazol und Voriconazol; (a.ii) ein Echinocandin ist, ausgewählt aus Anidulafungin, Caspofungin und Micafungin; (a.iii) ein Allylamin ist, ausgewählt aus Terbinafin, Butenafin, Amorolfen und Naftifin; (a.iv) ein Polyen ist, ausgewählt ist aus Amphotericin B und Nystatin; (a.v) ein Purin- oder Pyrimidinnukleotid-Hemmer ist, der Flucytosin ist; (a.vi) ein Mannan-Hemmer ist, der Pradamicin ist; (a.vii) ein Proteinelongationsfaktorhemmer ist, ausgewählt aus Sordarin und Analogen davon; oder (a.viii) ein Polyoxin ist, das Nikkomycin Z ist.
7. Pharmazeutische Zusammensetzung nach Anspruch 2, umfassend (i) eine Verbindung wie definiert in Anspruch 1, (ii) ein zweites antimykotisches Mittel wie definiert in einem der Ansprüche 3 bis 6, und (iii) einen oder mehrere pharmazeutisch annehmbare Träger und/oder Hilfsstoffe.
8. Verbindung nach Anspruch 1, Zusammensetzung nach Anspruch 2 oder 7 oder eine Kombination nach einem der Ansprüche 3 bis 6 zur Verwendung in einem Verfahren zur Behandlung des menschlichen oder tierischen Körpers durch Therapie.

9. Verbindung nach Anspruch 1, Zusammensetzung nach Anspruch 2 oder 7 oder eine Kombination nach einem der Ansprüche 3 bis 6 zur Verwendung in der Prävention oder Behandlung von Pilzerkrankung.
10. Verbindung, Zusammensetzung oder Kombination zur Verwendung nach Anspruch 9, wobei die Verbindung zur Verwendung in der Prävention oder Behandlung von Pilzerkrankung durch intravenöse Verabreichung ist.
11. Verbindung, Zusammensetzung oder Kombination zur Verwendung 8 oder Anspruch 9, wobei die Erkrankung durch eine Aspergillus-Spezies verursacht wird.
12. Verbindung, Zusammensetzung oder Kombination zur Verwendung 8 oder Anspruch 9, wobei die Erkrankung durch einen Hautpilz verursacht wird, oder wobei die Erkrankung allergische bronchopulmonale Aspergillose (ABPA) oder Asthma ist.
13. Kit, umfassend in Beimischung oder in separaten Behältern eine Verbindung wie definiert in Anspruch 1 und ein zweites antimykotisches Mittel wie definiert in einem der Ansprüche 3 bis 6.
14. Verfahren zum Kontrollieren einer Pilzerkrankung in einer Pflanze, wobei das Verfahren ein Anwenden einer Verbindung auf den Locus umfasst, die wie folgt ist:

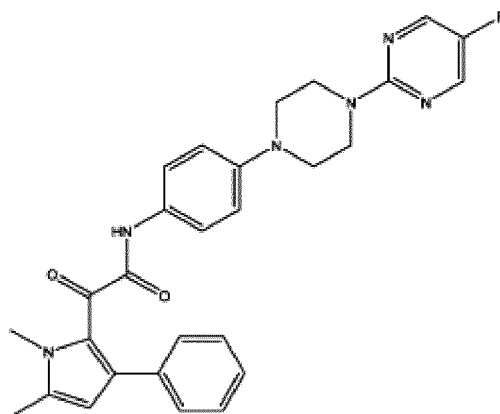
2-(1,5-Dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluorpyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamid oder ein landwirtschaftlich annehmbares Salz davon; und optional eines zweiten antimykotischen Mittels.

15. Verwendung einer Verbindung, wobei die Verbindung wie folgt ist:

2-(1,5-Dimethyl-3-phenyl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluorpyrimidin-2-yl)piperazin-1-yl)phenyl)-2-oxoacetamid oder ein landwirtschaftlich annehmbares Salz davon; optional zusammen mit einem zweiten antimykotischen Mittel als ein landwirtschaftliches Fungizid.

Revendications

1. Composé, lequel composé est :



du 2-(1,5-diméthyl-3-phényl-1H-pyrrol-2-yl)-N-(4-(4-(5-fluoropyrimidin-2-yl)pipérazin-1-yl)phényl)-2-oxoacétamide, ou un sel pharmaceutiquement acceptable de celui-ci.

2. Composition pharmaceutique comprenant un composé tel que défini dans la revendication 1, associé à un ou plusieurs vecteurs et/ou excipients pharmaceutiquement acceptables.
3. Combinaison pharmaceutique comprenant :

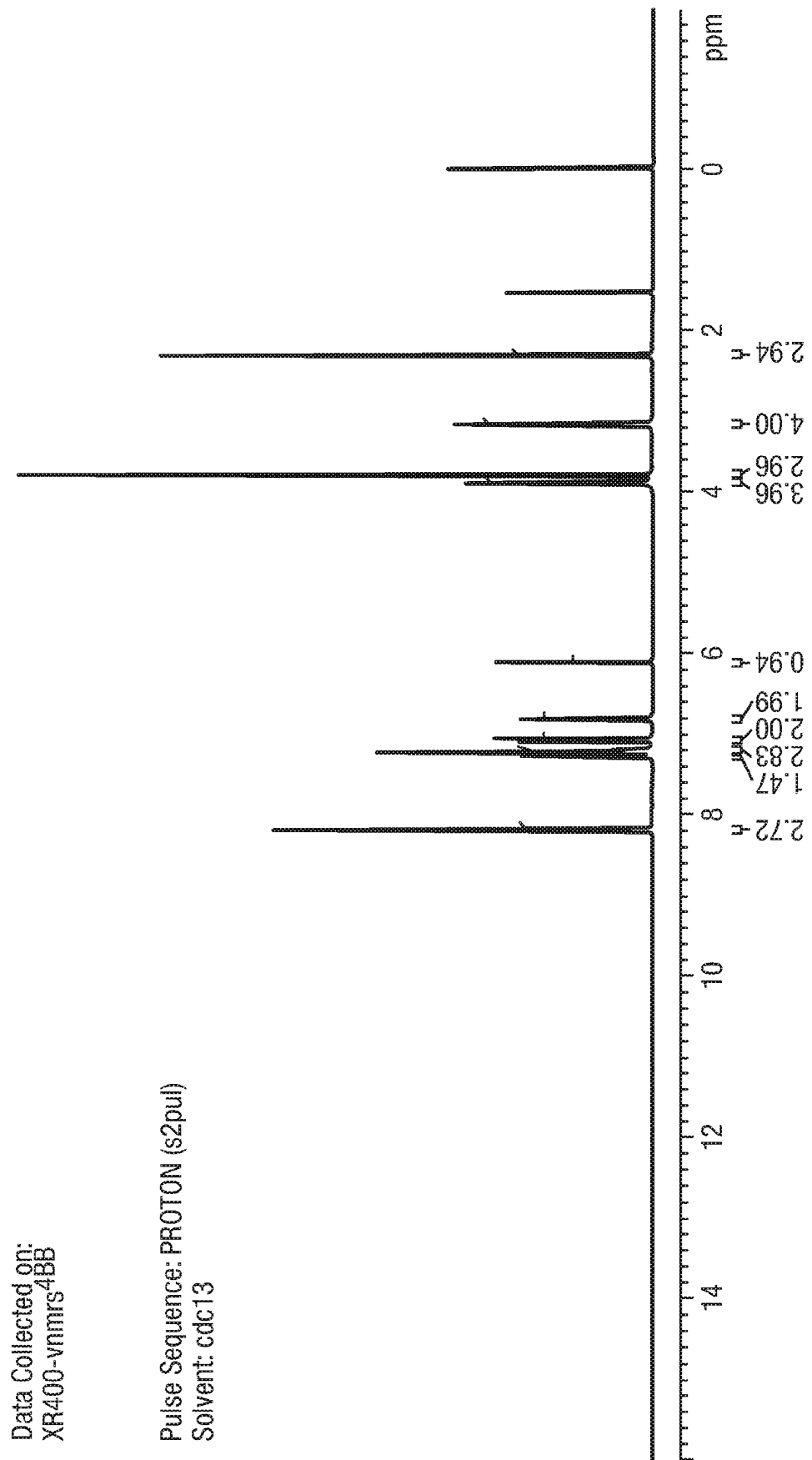


Fig. 1